

Total Internal Reflection - HKDSE Physics Summary

1. Introduction to Total Internal Reflection

1.1 Definition

Total Internal Reflection (TIR) occurs when light traveling in a denser medium hits the boundary with a less dense medium at an angle greater than the critical angle, and **all light is reflected back** into the denser medium with **no refraction**.

1.2 Key Conditions for TIR

For total internal reflection to occur, **BOTH** conditions must be satisfied:

1. ✓ Light travels from **denser to less dense** medium ($n_1 > n_2$)
2. ✓ Angle of incidence **greater than critical angle** ($i > c$)

If either condition is not met, TIR does NOT occur.

2. Critical Angle

2.1 Definition

Critical angle (c) is the angle of incidence in the denser medium for which the angle of refraction in the less dense medium is exactly 90° (refracted ray travels along the boundary).

2.2 Critical Angle Formula

$$\sin c = \frac{n_2}{n_1}$$

For light going from medium to air (

$$n_2 = 1$$

):

$$\sin c = \frac{1}{n}$$

or

$$c = \sin^{-1} \left(\frac{1}{n} \right)$$

Where:

- c
= critical angle
- n_1
= refractive index of denser medium
- n_2
= refractive index of less dense medium
- n
= refractive index of the denser medium (when other medium is air)

2.3 Common Critical Angles

Boundary	Refractive Index	Critical Angle
Water → Air	$n = 1.33$	$c \approx 49^\circ$
Glass → Air	$n = 1.50$	$c \approx 42^\circ$
Diamond → Air	$n = 2.42$	$c \approx 24^\circ$

Important:

- Higher refractive index → Smaller critical angle
- Diamond has smallest critical angle (most sparkle)

3. Three Cases at Boundary (Denser to Less Dense)

Case 1:

$$i < c$$

(Angle less than critical angle)

- **Most light is refracted** into less dense medium
- Refracted ray bends **away from normal**

- **Small amount reflected** back (partial reflection)
- **Normal refraction** occurs

Case 2:

$$i = c$$

(Angle equals critical angle)

- Refracted ray travels **along boundary** (
 $r = 90^\circ$
)
- Angle of refraction is exactly 90°
- This is the **critical condition**
- Defines the critical angle

Case 3:

$$i > c$$

(Angle greater than critical angle)

- **All light is reflected** back into denser medium
- **No refraction** occurs (no light escapes)
- **Total Internal Reflection** occurs
- Follows law of reflection:
 $i = r$

4. Comparison: Reflection vs Total Internal Reflection

Property	Ordinary Reflection	Total Internal Reflection
Surface	Any surface	Denser to less dense boundary
Light loss	Some light absorbed	No light loss (100% reflected)
Image quality	May be dim	Very bright, clear

Conditions	None	$n_1 > n_2$ and $i > c$
Efficiency	Lower	100% efficient

Advantage of TIR:

- No light loss (100% reflection)
- Brighter, clearer images
- Used in optical fibers, prisms

5. Applications of Total Internal Reflection

5.1 Optical Fibers

Structure:

- **Core:** High refractive index (carries light)
- **Cladding:** Lower refractive index (surrounds core)

How it works:

- Light enters fiber at small angle
- Hits core-cladding boundary at angle $>$ critical angle
- Total internal reflection keeps light trapped in core
- Light travels long distances with minimal loss

Applications:

- Telecommunications (internet, phone)
- Medical endoscopes (viewing inside body)
- Decorative lighting

5.2 Prisms in Optical Instruments

45°-45°-90° Prism:

- Light enters at 90° (no refraction)
- Hits internal surface at 45°

- For glass ($c \approx 42^\circ$): $45^\circ > 42^\circ$, so TIR occurs
- Light reflected by 90° or 180°

Applications:

- **Periscope:** Two prisms to see over obstacles
- **Binoculars:** Prisms to fold light path, make compact
- **SLR cameras:** Prism to direct light to viewfinder

Advantages over mirrors:

- No silvering needed (no degradation)
- 100% reflection (brighter image)
- More durable

5.3 Diamonds and Gemstones

Why diamonds sparkle:

- Very high refractive index ($n = 2.42$)
- Very small critical angle ($c \approx 24^\circ$)
- Light easily undergoes TIR inside diamond
- Multiple internal reflections before exiting
- Creates brilliance and "fire" (sparkle)

Cut design:

- Facets cut at specific angles
- Maximizes total internal reflection
- Enhances sparkle and brilliance

5.4 Mirages

Hot day mirage (road appears wet):

- Hot air near ground has lower density (lower n)

- Light from sky travels through layers of decreasing density
- Light bends away from normal at each layer
- Eventually reaches critical angle and undergoes TIR
- Reflected light appears to come from ground
- Creates illusion of water on road

Cold day mirage (objects appear elevated):

- Cold air near ground, warmer air above
- Light bends towards normal, then undergoes TIR
- Objects appear elevated or floating

5.5 Fish Eye View

Fish looking up from underwater:

- Light from above water can only enter within a cone
 - Cone angle determined by critical angle
 - For water (

$$c \approx 49^\circ$$
), cone angle $\approx 98^\circ$ ($2 \times 49^\circ$)
 - Outside this cone: TIR occurs, fish sees reflection of underwater objects
 - Fish sees entire above-water world compressed into this cone
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6. Problem-Solving Strategies

6.1 Determining if TIR Occurs

Checklist:

1. Is light traveling from denser to less dense? (

$$n_1 > n_2$$
)
2. Calculate critical angle:

$$\sin c = \frac{n_2}{n_1}$$
3. Compare angle of incidence with critical angle

4. If

$$i > c$$

: TIR occurs

5. If

$$i \leq c$$

: Normal refraction occurs

6.2 Critical Angle Calculations

Given refractive index, find critical angle:

1. Use

$$\sin c = \frac{1}{n}$$

(for medium to air)

2. Calculate

$$c = \sin^{-1} \left(\frac{1}{n} \right)$$

3. Check calculator in degree mode

Given critical angle, find refractive index:

1. Use

$$n = \frac{1}{\sin c}$$

2. Calculate and check if

$$n > 1$$

6.3 Ray Diagram for TIR

Steps:

1. Draw normal at boundary

2. Draw incident ray in denser medium

3. Check if

$$i > c$$

4. If yes: Draw reflected ray using

$$i = r$$

(law of reflection)

5. If no: Draw refracted ray bending away from normal

7. Important Concepts

7.1 Why TIR Only Works One Way

- TIR requires light from **denser to less dense**
- Light from less dense to denser: Always refracts (bends towards normal)
- Cannot have
 $i > c$
when going from less dense to denser
- Critical angle only exists for denser $\rightarrow$ less dense

7.2 Relationship Between n and c

$$n = \frac{1}{\sin c}$$

- Higher
 n
 $\rightarrow$ Smaller
 c
 $\rightarrow$ Easier to achieve TIR
- Lower
 n
 $\rightarrow$ Larger
 c
 $\rightarrow$ Harder to achieve TIR
- Diamond (high
 n
) sparkles more than glass (lower
 n
)

7.3 Energy Conservation in TIR

- In ordinary reflection: Some energy absorbed
- In TIR: **No energy loss** (100% reflected)
- Makes TIR very efficient for light transmission
- Basis for optical fiber technology

8. Common Misconceptions

8.1 Misconception 1

✗ **Wrong:** TIR can occur when light goes from less dense to denser medium

✓ **Correct:** TIR only occurs when light goes from **denser to less dense** medium

8.2 Misconception 2

✗ **Wrong:** TIR occurs at any angle of incidence

✓ **Correct:** TIR only occurs when

$$i > c$$

(angle greater than critical angle)

8.3 Misconception 3

✗ **Wrong:** Critical angle is the same for all materials

✓ **Correct:** Critical angle depends on refractive index; higher

$$n$$

→ smaller

$$c$$

8.4 Misconception 4

✗ **Wrong:** Some light refracts during TIR

✓ **Correct:** In TIR, **all light is reflected**, no refraction occurs (100% reflection)

8.5 Misconception 5

✗ **Wrong:** Optical fibers use ordinary reflection

✓ **Correct:** Optical fibers use **total internal reflection** (more efficient, no light loss)

9. Comparison Table: Three Cases

Condition	$i < c$	$i = c$	$i > c$
Name	Normal refraction	Critical angle	Total internal reflection

Refraction	Yes, bends away from normal	Yes, along boundary ($r = 90^\circ$)	No
Reflection	Partial (weak)	Partial	Total (100%)
Light escapes?	Yes (most)	Yes (along boundary)	No
Application	Lenses, prisms	Defines critical angle	Optical fibers, prisms

10. HKDSE Exam Key Points

10.1 Must Remember

1. **Two conditions for TIR:** Denser to less dense (

$$n_1 > n_2$$

) AND

$$i > c$$

2. **Critical angle formula:**

$$\sin c = \frac{1}{n}$$

(for medium to air)

3. **Three cases:**

$$i < c$$

(refraction),

$$i = c$$

(critical),

$$i > c$$

(TIR)

4. **TIR is 100% efficient:** No light loss
5. **Applications:** Optical fibers, prisms, diamonds, mirages
6. **Higher $n \rightarrow$ Smaller c :** Diamond sparkles more than glass

10.2 Common Question Types

1. **Determine if TIR occurs:** Check conditions

2. **Calculate critical angle:** Use

$$\sin c = \frac{1}{n}$$

3. **Calculate refractive index:** Use

$$n = \frac{1}{\sin c}$$

4. **Ray diagrams:** Draw TIR or refraction

5. **Applications:** Explain optical fibers, prisms, mirages

6. **Comparison:** TIR vs ordinary reflection

10.3 Exam Tips

- Always check **both conditions** for TIR
- Calculate critical angle **first** before determining TIR
- Draw neat ray diagrams with **normal**
- Label: Denser medium, less dense medium, angles
- State clearly: TIR occurs or does not occur
- Show all working in calculations
- Calculator in **degree mode**

11. Quick Memory Mnemonics

Two Conditions for TIR:

Dense to less, that's condition one
Angle greater than critical, then it's done

Critical Angle Formula:

$\sin c$ equals one over n
Higher n , smaller c , remember when

Three Cases:

Less than c , light refracts

Equal to c , along boundary acts

Greater than c , total reflection, that's the facts

Applications:

Fibers carry light so far

Prisms in binoculars are

Diamonds sparkle like a star

Mirages on roads from afar

TIR Advantages:

No light loss, hundred percent

Brighter image, that's the intent

Better than mirrors, no lament

Study Suggestions:

1. Memorize two conditions for TIR
2. Practice calculating critical angle
3. Understand three cases ($i < c$,
,
 $i = c$,
,
 $i > c$)
4. Learn applications: optical fibers, prisms, diamonds, mirages
5. Practice ray diagrams for TIR
6. Understand why TIR is more efficient than ordinary reflection
7. Remember relationship: Higher
 n
→ Smaller
 c